SURV/SURVMETH 720

Processing Issues in Surveys

Class #10

November 17, 2025

Last week…

- Discussion of different sources of measurement error
  - Interviewers (interviewer effects)
  - Respondents
  - Questionnaire/Instrument
  - Mode

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  - Interviewers (interviewer effects)
  - Respondents
  - Questionnaire/Instrument
  - Mode

Framing of question may affect responses

Schuldt et al. (2011)

You may have heard about the idea that the world's temperature may have been *going up* [*changing*] over the past 100 years, a phenomenon sometimes called '*global warming*' ['*climate change*']. What is your personal opinion regarding whether or not this has been happening? (Definitely **has not been** happening; Probably **has not been** happening; Unsure, but leaning toward it **has not been** happening; Not sure either way; Unsure, but leaning toward it **has been** happening; Probably **has been** happening; Definitely **has been** happening)[3]

Respondents also reported their political self-identification (Democrat, Republican, Independent, or Other/none of the above)

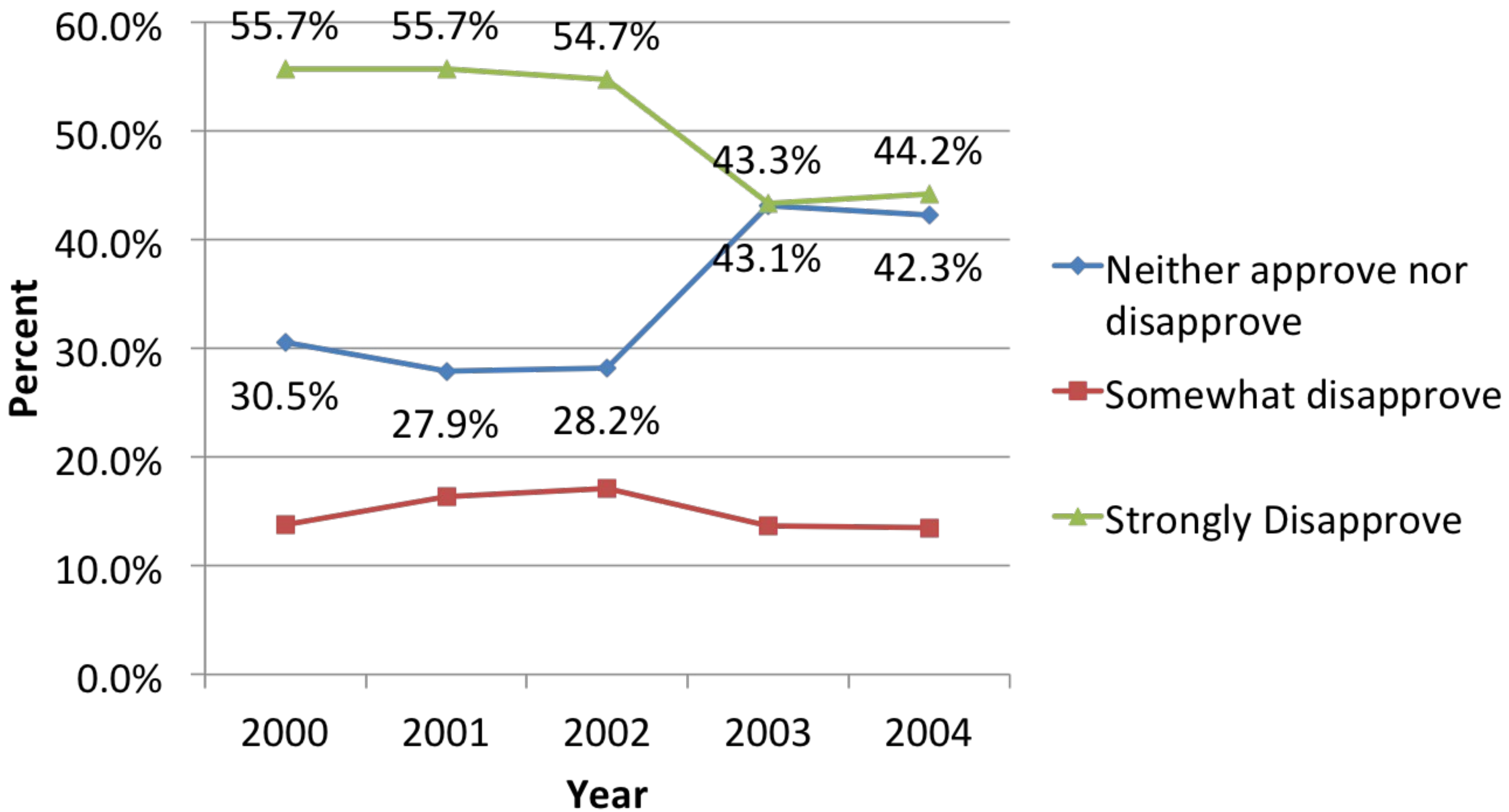
Table 2. Distribution of Existence Beliefs by Political Self-identification and Question Wording (GW = "global warming"; CC = "climate change")

	Overall		Republicans		Democrats		Independents		Others	
Reported Existence Belief	GW	CC	GW	CC	GW	CC	GW	CC	GW	CC
1 = Definitely has not been happening	6.6%	3.9%	12.7%	5.1%	1.9%	1.3%	6.6%	5.4%	4.7%	5.7%
2 = Probably has not been happening	8.4%	6.8%	18.2%	14.3%	2.6%	1.0%	6.2%	5.4%	3.7%	4.8%
3 = Leaning has not been happening	8.2%	7.6%	14.1%	11.4%	2.9%	5.2%	9.2%	6.6%	6.5%	5.7%
4 = Unsure	9.0%	7.6%	11.0%	8.9%	5.7%	6.0%	8.5%	8.7%	16.8%	6.7%
5 = Leaning has been happening	13.9%	15.7%	14.1%	17.0%	12.1%	13.6%	15.1%	15.7%	17.8%	18.1%
6 = Probably has been happening	25.6%	27.8%	20.2%	28.9%	29.2%	27.0%	27.9%	28.5%	23.4%	25.7%
7 = Definitely has been happening	28.2%	30.5%	9.7%	14.3%	45.6%	45.8%	26.5%	29.8%	27.1%	33.3%
% High Belief (≥ 5)	67.7%	74.0%	44.0%	60.2%	86.9%	86.4%	69.5%	74.0%	68.3%	77.1%
Mean Belief	5.05	5.30	3.95	4.62	5.94	5.94	5.09	5.29	5.18	5.37
N	1162	1099	362	370	421	382	272	242	107	105

Questionnaire – Question Order

- Preceding questions may affect responses
  - Assimilation effect: judgment of the current stimulus shifts in the direction of the preceding stimulus
    - “happiness with your health” -> “happiness with life as a whole”
  - Contrast effect: judgment shifts in the opposite direction
    - National Survey of Drug Use and Health: through 2002 order was “how do you feel about cigarettes” -> “how do you feel about marijuana”; from 2003 order was “how many times have you attacked someone” -> “how do you feel about marijuana”

Attitudes Towards Trying Marijuana



Wang, K., R. Baxter, and D. Painter. (2005). Modeling Context Effects in the National Survey of Drug Use and Health (NSDUH). Proceedings of the Joint Statistical Meetings.

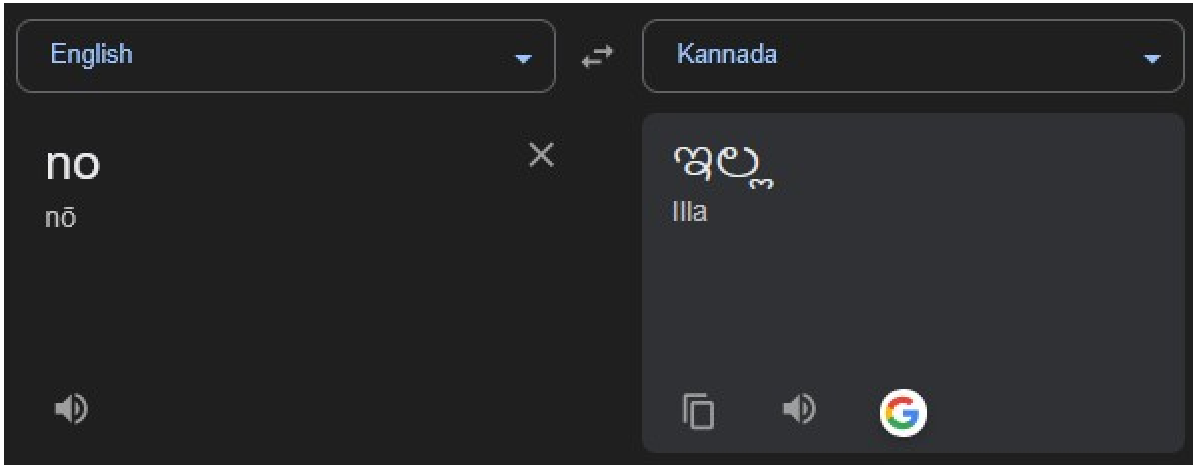
Good news

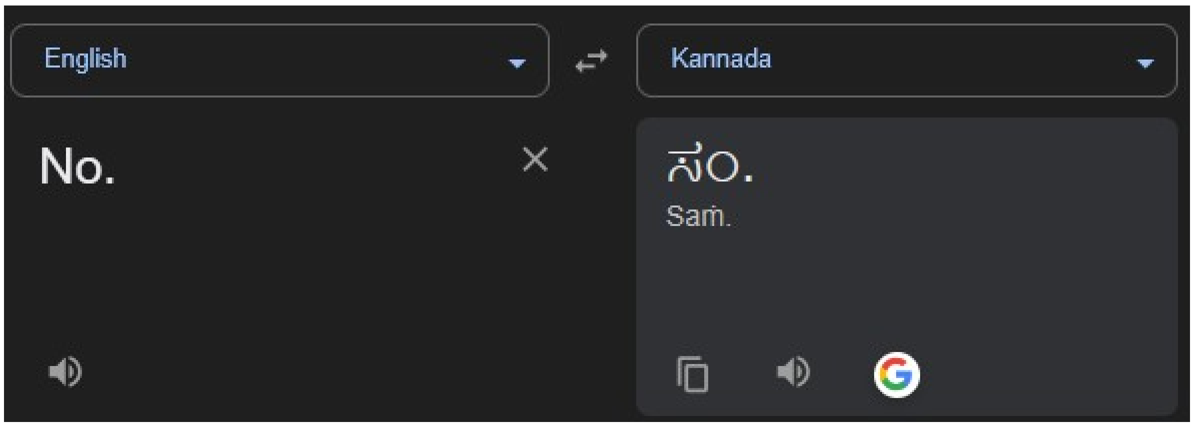
Chen et al. (2021) examine 110 experiments manipulating response order and wording. They find that:

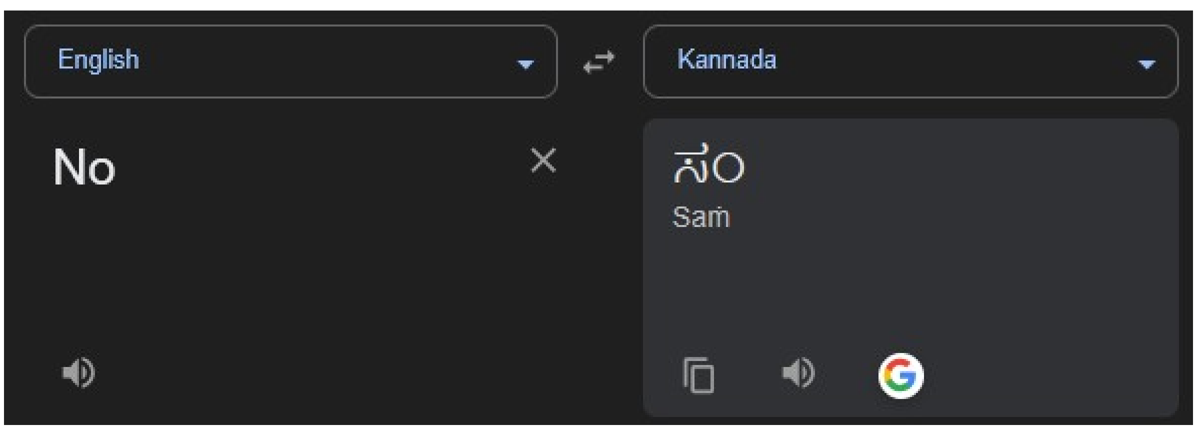
"…results are consistent with the conclusion that survey results on climate change issues are relatively robust, so policymakers can take them seriously if they wish to do so."

Before finishing this module..

IHDS example

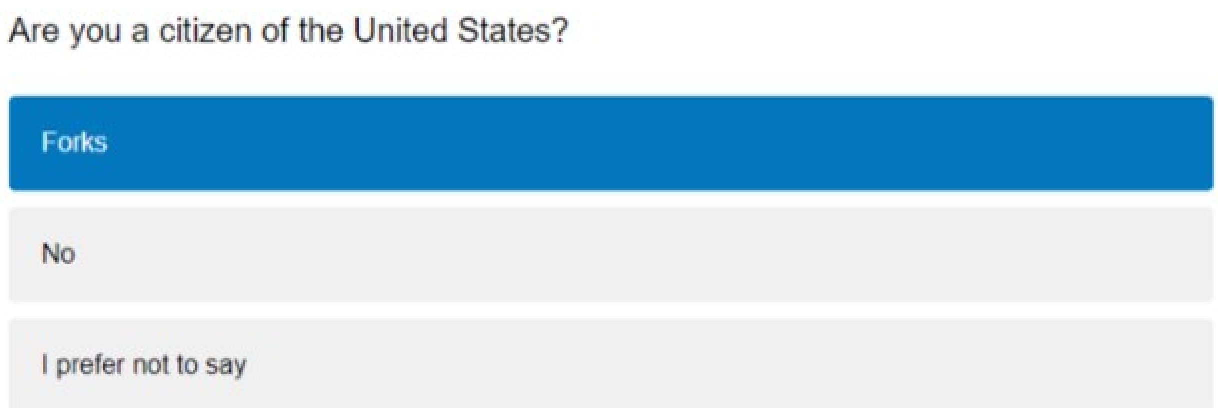






Interprets "No" as #

Are you a citizen of the United States?



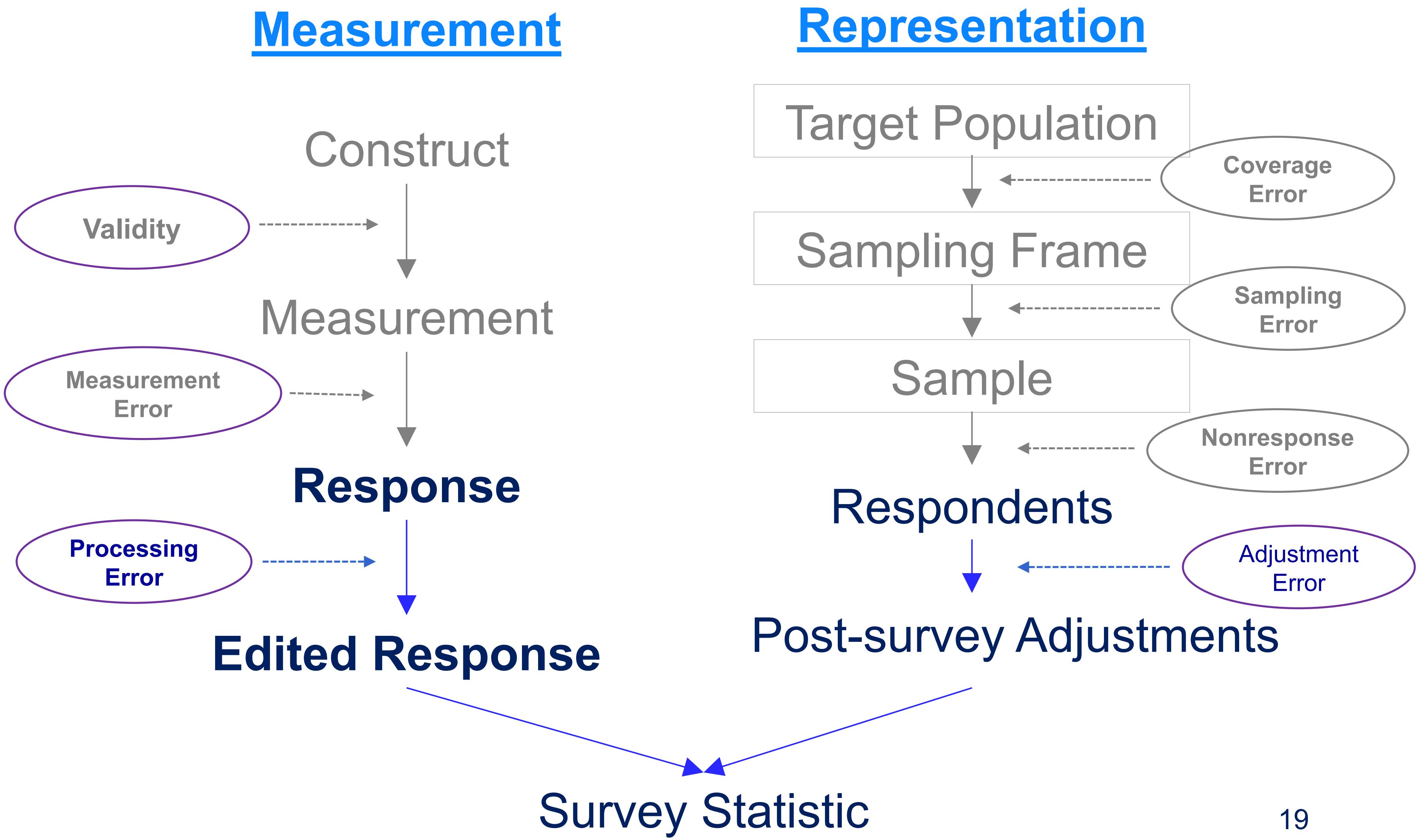
https://www.pewresearch.org/decoded/2025/03/21/how-a-glitch-in-an-online-survey-replaced-the-word-yes-with-forks/

Processing error

Data Processing

- Set of activities aimed at converting the survey data from its raw state as the output from data collection to a cleaned and corrected state that can be used in analysis, presentation, and dissemination (Biemer and Lyberg, 2003).
- Can have lot of resources committed to it.
- One important purpose is to correct errors. But errors can arise during the process (analogous to adjustment error)
  - Focus is typically on errors at the respondent level, not the estimate level
- Relatively little research on processing error compared to other errors.

TSE Framework



Data Processing

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- Can have lot of resources committed to it (possibly relative more in business/establishment surveys)

Data Processing…

Traditionally, the goal of editing has been to discover all errors in data and to “correct” them. We now realise that editing can result in errors (Granquist and Kovar(1997))

http://gauss.stat.su.se/master/statdatabaser/HT10/Literature/SwedishEditingMethods.pdf

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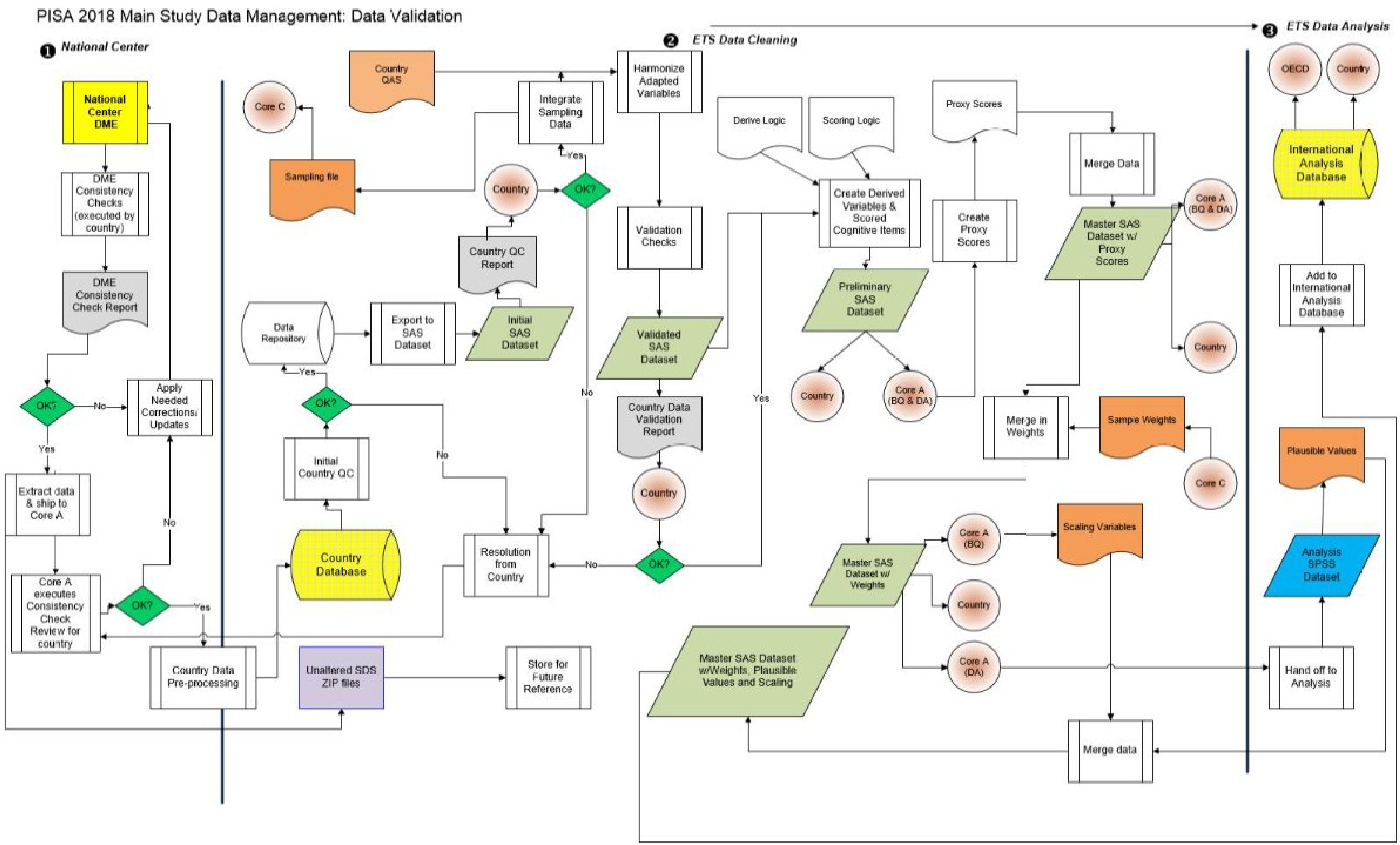
- Relatively little research on processing error compared to other errors.

Impression : this is “for the ops folks”, …



Source: https://www.oecd.org/pisa/data/pisa2018technicalreport/PISA2018%20TecReport-Ch-10-Data-Management.pdf

Three sources of processing error (among others)

1. Data entry

2. Editing
   – correcting errors
   – disclosure avoidance (maybe)

3. Coding open responses

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Data Entry/Capture

- Two main activities for converting responses into computer-readable format
  - **Human keying** by interviewers during data collection (include here?) or by keyers at central processing center after the survey (for paper questionnaires)
  - **Scanning** of hand-written marks and characters on paper questionnaires at central processing center
- Error rates are typically small
  - May depend on keyer skills, quality control process, and legibility of hand-writing
  - In principle, such errors are eliminated in web surveys

Three sources of processing error (among others)

1. Data entry

2. **Editing**
   - **correcting errors**
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3. Coding open responses

Editing approaches

- “Accountant’s view”
  - Exhaustive search for errors
- Practical: Significance editing/Selective editing
  - Focus on “significant” errors

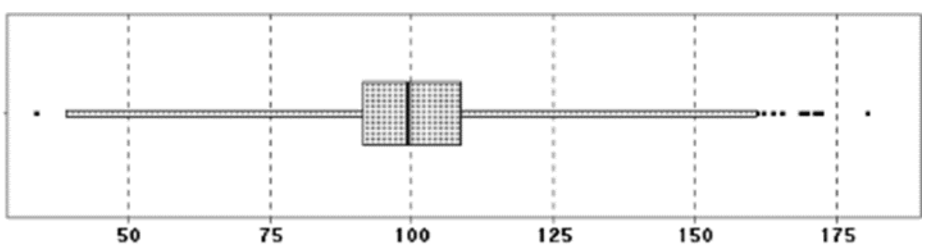
“fatal” vs “suspected” errors

Table 10.1: Range restriction rules for inconsistent and extreme values in the student file

Sequence	Description	SAS Code
1	Invalidate if number for an individual's weight is negative.	if (WB151Q01HA < 0) then WB151Q01HA=.I;
2	Invalidate if number for an individual's height is negative.	if (WB152Q01HA < 0) then WB152Q01HA=.I;
3	Invalidate if number of class periods per week in test language lessons (ST059Q01TA) is greater than 40.	if (ST059Q01TA > 40) then ST059Q01TA =.I;
4	Invalidate if number of class periods per week in maths (ST059Q02TA) is greater than 40.	if (ST059Q02TA > 40) then ST059Q02TA =.I;
5	Invalidate if number of class periods per week in science (ST059Q03TA) is greater than 40.	if (ST059Q03TA > 40) then ST059Q03TA =.I;
6	Invalidate if number of <class periods> per week in foreign language is greater than 40.	if (ST059Q04HA > 40) then ST059Q04HA= .I;
7	Invalidate if number of total class periods in a week (ST060Q01NA) is greater than 120 or less than 10	if (ST060Q01NA > 120 or ST060Q01NA < 10) and NOT MISSING(ST060Q01NA) then ST060Q01NA =.I;

Consistency checks example from Programme for International Student Assessment (2018).

https://www.oecd.org/pisa/data/pisa2018technicalreport/PISA2018%20TecReport-Ch-10-Data-Management.pdf



Number of students in a grade

What is an acceptable range?

Editing to Correct Errors

Two main steps:

- **Detection** : identify erroneous or missing values
- **Correction** : estimate new values

Detection Techniques

- Apply **edit rules** that describe admissible values for variable, often in relation to other variables
  - Examples: range, ratio, and consistency edits
  - Computerized instruments can do this in real-time [not really a *post-survey* edit] in a limited way
    [and may not be perfect: IHDS example]
  - Establishment surveys tend to have large number of rules

Detection Techniques

- Apply **edit rules** that describe admissible values for variable, often in relation to other variables
  - Examples: range, ratio, and consistency edits
  - Computerized instruments can do this in real-time [not really a *post-survey* edit] in a limited way
  - Establishment surveys tend to have large number of rules
- Identify and manually review **influential values** that have substantial impact on the estimates
  - checks of highest and lowest values, outlier detection
- Review number of key aggregate estimates. Identify **unusual estimates** and manually review individual values used to compute the estimate

Correction Techniques

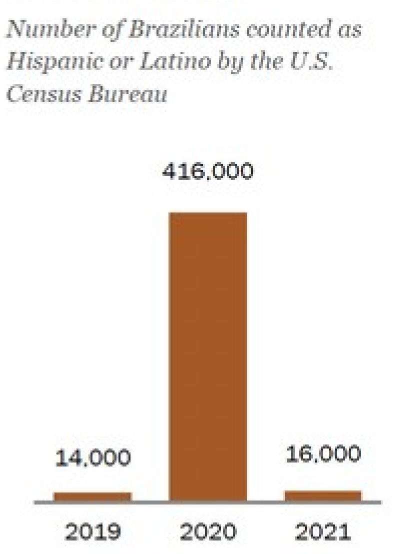
- **Manual correction**: respondent is recontacted or subject-matter expert is used to get a new value
  - Other edit rules can also be applied
- **Imputation**: used automated approach to assign new value (e.g., prior wave’s value, hot deck, regression,…)

Error Considerations

In principle, errors are corrected by this process. But error could arise for different reasons:

- Several values are inconsistent but the wrong one(s) is/are identified as erroneous
- Bad edit checks (perhaps created by people without knowledge of the concepts being measured)
- "Influential" answers that are actually correct
- Imputation error
  - using prior wave's value when actual circumstances have changed; omitting a key variable; modeling errors, etc.

ACS coding error



https://www.pewresearch.org/short-reads/2023/04/19/how-a-coding-error-provided-a-rare-glimpse-into-latino-identity-among-brazilians-in-the-u-s/

- Officially, Brazilians are not considered to be Hispanic or Latino -> in most cases people who report their Hispanic or Latino ethnicity as Brazilian in Census Bureau surveys are later recategorized – or "back coded" – as *not Hispanic or Latino*.

- 2020 ACS -> Census Bureau inadvertently left Brazilians and some other groups out of its back-coding procedures

- Side-note: being Hispanic or Latino means different things to different people

Editing Considerations

- Challenges
  - Costs: to 20% in household surveys and up 40% in federal establishment surveys (Granquist & Kovar, 1997)
  - While costs have come down, a significant proportion of the budget can still be devoted to editing

Table 1.1 Average proportions of the costs of sub-processes in the total cost of statistical surveys during 2003 and 2004.

Process	Proportion of total cost		
	All surveys	Short-period	Annual surveys and periodic
Respondent service	3.3	3.3	3.4
Manual data review	4.4	3.9	5.1
Data-registration editing	5.6	5.1	6.5
Production editing	15.3	12.7	18.9
Output editing	3.9	3.4	4.8
Total editing cost	32.6	28.3	38.6

http://gauss.stat.su.se/master/statdatabaser/HT10/Literature/SwedishEditingMethods.pdf

Editing Considerations

- Challenges
  - Costs

- Overediting
  - Imposing ones ideas about ‘correct data’
  - Too many minor changes: Reducing the usual editing effort by about half could reduce this difference to less than one percent at the global level and less than 10 percent at the detailed level (Granquist & Kovar, 1997)

Finally, one must not overlook the indirect costs associated with an undue degree of confidence in data quality and respondent reporting capacity that result from overediting the survey data sets. This is of particular importance when, as is usually the case, performance measures and audit trails are scarce or non-existent. Overedited survey data will often lead analysts to rediscover the editors' models, generally with an undue degree of confidence. For example, it was not until a demographer "discovered" that wives are on average two years younger than their husbands that the edit rule which performed this exact imputation was removed from the Canadian Census system!

Granquist and Kovar 1997

Editing Considerations

- Challenges
  - Costs
- Overediting
- Editing error rarely estimated and rarely reported in technical reports.
- Opportunity: Use edits to learn about measurement process and inform revisions

Data Flags

Reported Data	
C	Without contacting the respondent, the analyst corrects data provided in an inappropriate item code (functional category) or individual unit, corrects data so detail adds to total, corrects other reported minor reporting errors that prevents the use of the original data provided by the respondent.
K	Analyst corrects improperly keyed data and replaces with the reported values from the questionnaire.
R	1) Data item is reported directly by the respondent. 2) Respondent does not complete the survey form, but provides additional information which is compiled and used to complete the form, i.e., annual report, Website, etc.
T	Respondent reports totals and these data are prorated based on the prior year distribution.
U	Analyst obtains correct data from the respondent via telephone, e-mail, or fax.
V	1) Analyst verifies the data with the respondent via telephone, e-mail, or fax. 2) Analyst corrects improperly keyed data and replaces with the reported values from the questionnaire.

https://www.census.gov/programs-surveys/apes/technical-documentation/code-lists/data-flags.html

Imputation flag for CERTVOC (F_CERTVOC)

Imputation Flags

Type Imputation Flag

Imputation flag for CERTVOC

Category	Label	Frequency UnW[1]	Frequency W[2]	Percent UnW[1]	Percent W[2]
0	0 Reported Value	45,108	187,247,691.18	94.48	95.37
3	3 Donor Imputed	2,636	9,094,850.11	5.52	4.63
TOTAL		47,744	196,342,541.29	100.00	100.00

[1] UnW = Unweighted

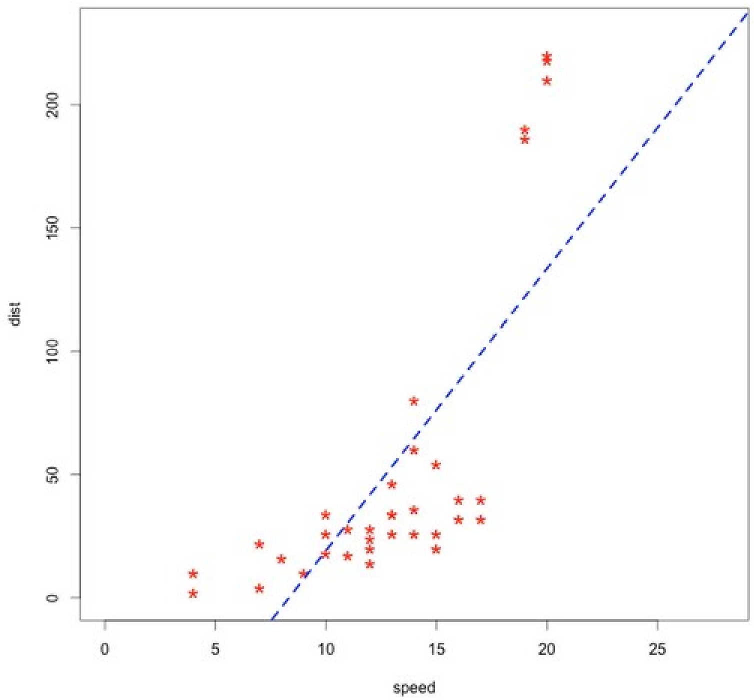
[2] W = Weighted

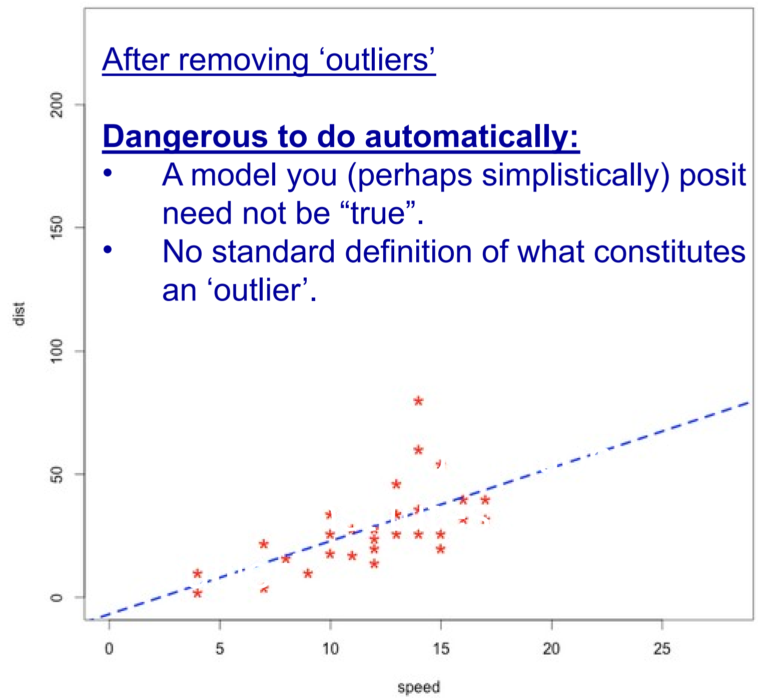
Note: Frequency values displayed here are calculated using the entire dataset. Frequency values displayed in your statistical software may be calculated based on your chosen population of analysis.

https://nces.ed.gov/datalab/

Finally, don’t forget about editing due to ‘outliers’

- May not be errors at all !
- Dangerous to remove such data points in an automatic fashion





After removing ‘outliers’

Dangerous to do automatically:

- A model you (perhaps simplistically) posit need not be “true”.
- No standard definition of what constitutes an ‘outlier’.

Plot source: r-statistics.co/Outlier-Treatment-With-R.html

A lesson from a non-survey setting

Why hadn't they discovered the phenomenon earlier? Unfortunately, the TOMS data analysis software had been programmed to flag and set aside data points that deviated greatly from expected measurements and so the initial measurements that should have set off alarms were simply overlooked. In short, the TOMS team failed to detect the ozone depletion years earlier because it was much more severe than scientists expected.

https://earthobservatory.nasa.gov/features/RemoteSensingAtmosphere/remote_sensing5.php

Three sources of processing error (among others)

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3. Coding open responses

Editing for Disclosure Avoidance

- Goal is to protect confidentiality of respondents by reducing the probabilities that they can be identified in the released data
- Two basic approaches
  - Microdata edits
  - Macrodata edits

Techniques for Editing Microdata

- Removing personally identifying information (PII)
  - names, addresses, ID numbers
- Coarsen the original data
  - top coding, recoding into intervals, limiting geographical detail
- Data swapping: switching the data values for selected records with those for other records
- Adding noise: changing original values in a random way
- Partially synthetic data: sensitive variables are blanked out and replaced with imputed values

Techniques for Editing Macrodata

- Combining tables cells to form larger groupings.
- Changing values in systematic way (rounding).

Quality Considerations

- In principle, estimates based on the original data and altered data should be similar. But there are quality considerations:
  - Too many alterations to data (e.g., all imputed values) may decrease its usefulness
  - As noise is added to microdata, variance may increase
  - Less detailed reports may be less useful to data users
- “secondary data analysts act as if the released data are in fact the collected sample, thereby ignoring any inaccuracies that might result from the disclosure treatment” – Reiter 2012

Three sources of processing error (among others)

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Seems innocuous but…

Many users believe that open-ended coding is easy to do, that it generates valid measures, and that it's performed well by survey organizations. The evidence tells a different story.

- Lupia 2018

Researchers are often not aware of the complex preparatory work behind occupational coding, and often consider the published variable of 'occupation' as free of error. This is not the case if processing errors arise during the coding of the variable.

– Belloni 2016

Coding Open Response

- Unlike closed responses that are automatically stored as numeric values, open questions need to be coded into categories after the survey.

- Coding process may depend on type of open questions:
  A. Open items that solicit unconstrained answers
  B. Open items that imply a constrained set of answers

A. Open items that solicit unconstrained answers

- Invites the respondent to express their answer with no constraints on its structure or length
- Example from the American National Election Survey : "What do you think are the most important problems facing this country?"
- Answer can be long and cover more than one topic, making it difficult to "field code" by interviewer in real-time
- **Manual coding** (after data collection) requires that coders read each individual response and group it into a small number of categories. For example, the categories used for the above question include: "Crime"; "Race relations"; "Immigration"; "Size of government" and "Poverty".

B. Open items that imply a constrained set of answers

- Constrained set of answer categories are implied
- Examples: current medications, questions about one's occupation, work duties, hobbies,
- Answers may be relatively short (e.g., 1 or 2 words) and classified into a single category
- Possible coding approaches
  - Field-coding by interviewers and verification from respondent
  - Automated coding after the survey
  - Emerging methods: in-survey coding
  - AI, AI, AI, …..

Coding Error - Bias

- In principle, this will occur when code assigned does not reflect “true value”.

- Hard to sometimes quantify: may not be a single correct answer.

Coding Error - Variance

- Occurs when coders do not assign same code
- Intercoder reliability is the typical measure of coding quality
  - easy to compute
  - percent agreement is the single most widely used index (tends to be higher when there are fewer categories because of chance agreement)
  - other measures such as “Cohen's kappa”

Design Effect for Coders

- Inflation of the variance due to coders

$$\text{Deff}_{coder} = 1 + \rho_c(m - 1)$$

ρ_c = intra-class correlation for coders

m = average workload

- Analogous to deff_{int}
- The more variance due to coders, the less precise the estimate.
- Campanelli (1997): Finds deff's of 1 – 1.79 with workloads of 300-400.

Potential reasons for coding error (reasons for coder disagreement)

- Response issues
  - Ambiguous response, not probed enough
  - Erroneously recorded responses, e.g., misspelled or omitted words

- Ambiguity of correct code number
  - Variation in judgments. Coders may disagree based on their interpretation of the coding instructions and survey response.

Potential reasons for coding error…

- Conflicting/Ambiguous coding rules
  - "If you see "assistant anything," drop the "assistant" and code to the other word" v/s:
  - If you see "teacher's assistant," drop the "teacher" and code "assistant."
- Coding rules not properly applied
- Coding error is not just about being inaccurate due to the coder:
  - Often no best category that fits a response
  - A response may fit more than one category to different degrees

How to reduce coding error?

- Formal documentation
- Training is needed but coders' knowledge and experience count a lot – so get new coders to code a lot !
- Obtain a set of responses prone to disagreement – get coders to discuss.
- Workload ?
- Focus on getting it right at first rather than "my sup will catch it".

Also earlier…

- Train interviewers to probe when required (and not over-probe/over-record)
  - In general, longer occupation descriptions more prone to be less reliably coded ! (Conrad, Couper, and Sakshaug 2016)
    - ➢ But for difficult occupation terms, longer descriptions were slightly more reliably coded than short descriptions

  (Conrad et al., 2016; Massing et al, 2019)

What about validity ?

- Most discussion around inter-coder reliability.

- But high reliability can be due to all coders being equally lax or poor instructions

- Validity of the coding process depends on the quality of coding instructions, whereas inter-coder reliability depends on the implementation of these instructions (Hak and Bernts 1996)

- "coders rely on the use of arbitrary coding rules based on superficial features of the description" (Conrad et al. 2016)

  Also might explain low inter-agency reliability (Massing et al., 2019)

From Lupia, 2018:

> Now we have a set of questions concerning various public figures. We want to see how much information about them gets out to the public from television, newspapers and the like.... What about...William Rehnquist – What job or political office does he NOW hold?

- Only 12% provided a “correct” response
  - The verdict is stunningly, depressingly clear: most people know very little about politics – Luskin, 2002
- But what is correct ? Only if respondents specifically identified that Rehnquist was “Chief Justice” **and** on the “Supreme Court”
- 30 percent of respondents that identified him as a Supreme Court justice were marked “incorrect”. This would be “incorrect”: “Supreme Court justice. The main one.”
- Can look at validity by looking at correlation with expert evaluation (see Campenelli et al., 1997)

Occupation coding

- Stand-out example of coding open-ended responses:
- Three methods:
  - Fully manual
  - Fully automatic
  - Computer-assisted

Manual coding

- Free text responses assigned codes by coders assign codes based on a standardized classification system
- Encourage coders to be explicit about uncertainties (e.g., assign a secondary code possibility)
  - The coding scheme itself can help here, e.g., International Standard Classification of Occupations 1988 (ISCO-88) has a hierarchical structure
  - Higher level codes used in case of lack of information at lower levels.

I. Major groups

1 Managers
2 Professionals
3 Technicians and Associate Professionals
4 Clerical Support Workers
5 Services and Sales Workers
6 Skilled Agricultural, Forestry and Fishery Workers
7 Craft and Related Trades Workers
8 Plant and Machine Operators and Assemblers
9 Elementary Occupations
0 Armed Forces Occupations

II. Major and Sub-major groups

1 Managers
- 11 Chief Executives, Senior Officials and Legislators
- 12 Administrative and Commercial Managers
- 13 Production and Specialized Services Managers
- 14 Hospitality, Retail and Other Services Managers

III. Major, Sub-major, and Minor groups

- 14 Hospitality, Retail and Other Services Managers
  - 141 Hotel and Restaurant Managers
  - 142 Retail and Wholesale Trade Managers
  - 143 Other Services Managers

IV. Major, Sub-major, Minor, and Unit groups

- 14 Hospitality, Retail and Other Services Managers
  - 141 Hotel and Restaurant Managers
    - 1411 Hotel Managers
    - 1412 Restaurant Managers
  - 142 Retail and Wholesale Trade Managers
    - 1420 Retail and Wholesale Trade Managers
  - 143 Other Services Managers
    - 1431 Sports, Recreation and Cultural Centre Managers
    - 1439 Services Managers Not Elsewhere Classified

Belloni et al (2016) find high incidence of coding error at even the one-digit level (28% for "last job"). Agreement rates are 55% - 80% at a 3-digit level (Elias, 1997; Mannetje and Kromhout, 2003)

2. Fully automatic/Computer-assisted methods

- As the ISCO example showed, coding can get complicated + manual coding is expensive and time-consuming (especially for large surveys).

- Automatic methods are attractive: based on either a simple look-up table based on historical data or more sophisticated statistical methods or using LLMs

Examples:

- NIOSH Industry and Occupation Computerized Coding System (NIOCCS) https://csams.cdc.gov/nioccs/Default.aspx

- labourR – R package

- CASCOT: Warwick Institute for Employment Research https://warwick.ac.uk/fac/soc/ier/software/cascot/details/

CASCOT: Computer Assisted Structured Coding Tool

- Can run in three different modes: fully automatic, semiautomatic, or manual/one-by-one

- Assigns a certainty score (1 - 100). Common practice to accept automatic coding if the score is greater than, say, 70 (e.g. Belloni et al 2016)

- Software suggests alternate codes – for low scores, manual coders can decide based on these suggestions.

- CASCOT has been compared to high quality manually coded data: 80% records receive a score > 40 and of these, 80% matched the manually coded data.

Fully automatic/Computer-assisted methods…

- **Supervised** machine learning approaches try to replicate manual-coded results. Basic process:
  1. split into two parts (training and test)
  2. manually code the training data
  3. fit model to training data
  4. use model to predict categories in test data
- **Unsupervised** machine learning try to discover new coding categories/themes
  - e.g., topic modeling, e.g., Roberts et al 2014: “Structural topic models for open-ended survey responses.”

Hybrid approach may reduce costs while maintaining acceptable levels of accuracy, see Schonlau and Couper (2016)

Occupation coding during the interview

Schierholz et al, 2019

- Interviewer then runs a query (supervised algorithm) which suggests categories that the R can choose from
- Algorithm suggested a category in 90% of cases (10% mostly due to misspelt words). 72% Rs selected a category from the automatically generated categories (reduces cost of manual coding)
- Does not increase IW duration
- Additional probing questions were not useful
- One advantage of this approach: the interviewer knows when to probe.

Using AI for coding open-ended resonses

- But also to: probe when an insufficient open-ended response received
  - Sturgis, 2025 (AAPOR 2025): https://tinyurl.com/LSEps25 (~80% agreement with manual coding)
  - Geisen et al. (2025). Respondents seem to find follow-ups more engaging.

Some general directions

- “Few shot” > “Zero shot”
- Pretraining necessary
- Prompts important. Ideally provide both positive and negative examples.
- Performance differs between LLM/NLP models
- Gweon and Schonlau, 2024:
  - BERT for coding open-ended responses
  - Find that fine-tuning the model is essential
  - If manually trained on 100 observations, BERT barely beats non-pretrained machine learning approaches
  - Advantage rapidly increases with more coded observations (e.g., 200-400)

What do you think and likely challenges and pitfalls in using AI for coding open-ended responses?

Next Week

- Multiple Sources of Survey Error
- Reading assignment
- Questions?